

Remote Patient Vitals Alert & Monitoring App (RPVAMS)

Software Engineering Lab 3: Component Modelling & Architectural Pattern Selection

Course: UE24CS341A - Software Engineering

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Scenario No.: Scenario 16 (Healthcare & Telemedicine)

Primary Target Actors: Remote Patient, On-Call Caregiver, Backup Caregiver

1. Introduction & Problem Statement

In continuation of Lab 1 (Requirements Engineering & UML Use-Case Modelling), this specification defines the Component Architecture and Architectural Pattern Selection for the Remote Patient Vitals Alert & Monitoring App (RPVAMS).

Post-operative home-care patients are at acute risk of rapid hemodynamic instability, hypoxia, and cardiac decompensation. The RPVAMS delivers a resilient, high-throughput continuous telemetry ingestion pipeline that captures biometric data (SpO2, Heart Rate, Blood Pressure) from wearable medical sensors, evaluates metrics against individualized clinical baselines in under 2 seconds, and orchestrates automated emergency alert escalations across an on-call caregiver matrix.

1.1 Core Requirements Realized in Architecture

- FR-001 (Real-Time Ingestion & Evaluation): The system shall continuously evaluate ingested patient vitals against configured clinical thresholds and flag critical anomalies within 2 seconds.
- FR-002 (Authenticated Live Dashboard): The system shall allow a Remote Patient and On-Call Caregiver to authenticate securely and view the live vitals dashboard displaying continuous SpO2, Heart Rate, and Blood Pressure streams.
- FR-003 (Critical Alert Dispatch & Acknowledgment): The system shall dispatch immediate critical alerts to the On-Call Caregiver upon detecting a vital threshold breach and record caregiver acknowledgment.
- FR-004 (Dynamic Threshold Configuration): The system shall allow an On-Call Caregiver to configure and dynamically update clinical baseline thresholds for patient vitals without system downtime.
- FR-005 (Caregiver Matrix Timeout Escalation): The system shall automatically escalate unacknowledged critical alerts to a Backup Caregiver if the primary On-Call Caregiver does not acknowledge within the 60-second SLA.

- NFR-001 (Scalability & Uptime): The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime and sub-second ingestion latency.
- NFR-002 (Security & Compliance): The system shall enforce end-to-end encryption using TLS 1.3 in transit and AES-256 at rest, coupled with strict Role-Based Access Control (RBAC) and immutable audit logging.

2. UML 2.5 Component Modelling Specification

The RPVAMS architecture is structured into four cohesive, decoupled subsystems: Client Presentation Subsystem, Secure Edge Gateway Subsystem, Core Application & Processing Subsystem, and Data Persistence & Storage Subsystem, supported by a cross-cutting Compliance & Audit Subsystem. The design utilizes official UML 2.5 notation with <<component>> stereotypes, explicit provided interfaces (ball/lollipop notation), required interfaces (socket notation), ports, and database stereotypes.

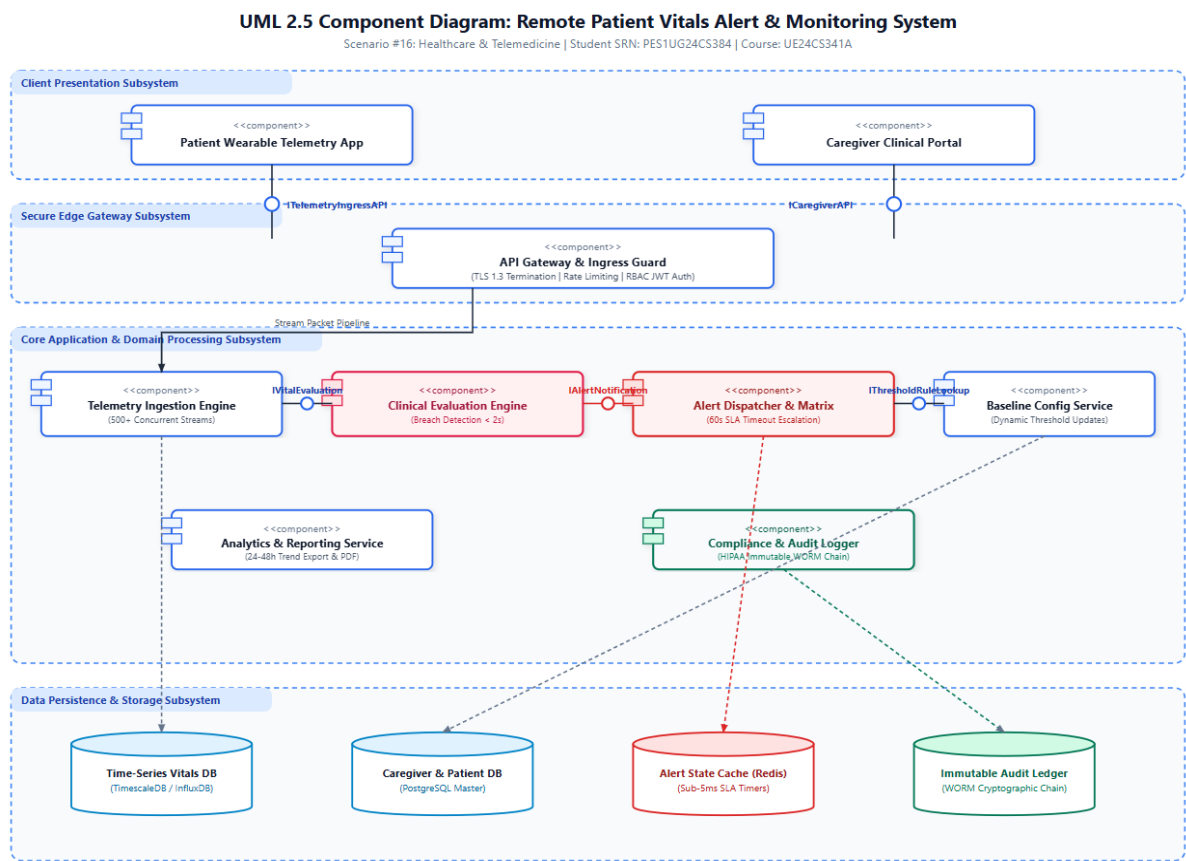


Figure 1: UML 2.5 Component Diagram of the Remote Patient Vitals Alert & Monitoring System

3. Detailed Component & Interface Catalog

Component Name	Tier / Subsystem	Responsibilities	Provided / Required Interfaces	Traceability
Patient Wearable Telemetry App	Client Presentation	Mobile/wearable client streaming encrypted biometrics, rendering personal vital dashboard, and displaying active caregiver status.	Provided: IPatientTelemetryAPI Required: ITelemetryIngressAPI, IAuthAPI	FR-001, FR-002
Caregiver Clinical Portal	Client Presentation	Web/Mobile responsive portal for real-time telemetry monitoring, emergency alert sirens, one-touch triage acknowledgment, and threshold configuration.	Provided: ICaregiverPortalAPI Required: ICaregiverAPI, IAuthAPI	FR-002, FR-003, FR-004, FR-005
API Gateway & Ingress Guard	Secure Edge & Perimeter	Performs TLS 1.3 termination, rate limiting, DDoS defense, token-bucket traffic shaping, and RSA-256 JWT RBAC validation.	Provided: IEdgeGatewayAPI, ITelemetryIngressAPI, ICaregiverAPI Required: IAuthService, ITelemetryStream	NFR-001, NFR-002
Telemetry Ingestion Engine	Core Application Tier	High-throughput asynchronous stream ingestion gateway parsing SpO2, HR, and BP telemetry payloads across 500+ concurrent channels.	Provided: ITelemetryStream Required: IVitalEvaluation, ITimeSeriesStore, IAuditLog	FR-001, NFR-001
Clinical Evaluation Engine	Core Application Tier	Real-time stream processing engine evaluating biometrics against dynamic patient baselines within 2s and classifying anomaly severity.	Provided: IVitalEvaluation Required: IThresholdRuleLookup, IAlertNotification, IAuditLog	FR-001, NFR-002
Alert Dispatcher & Escalation Matrix	Core Application Tier	Manages emergency alert distribution, tracks 60s caregiver acknowledgment SLA via Redis state machine, and auto-	Provided: IAlertNotification, IAlertAckService Required: IEscalationSLA,	FR-003, FR-005

		escalates to Backup Caregivers.	INotificationService, IAuditLog	
Baseline Config Service	Core Application Tier	Maintains patient clinical profiles and provides non-disruptive dynamic threshold calibration for attending healthcare providers.	Provided: IConfigService, IThresholdRuleLookup Required: IProfileDataStore, IAuditLog	FR-004
Analytics & Reporting Service	Core Application Tier	Aggregates historical biometrics over 24-48h windows and renders downloadable clinical trend analytics reports (PDF/CSV).	Provided: IAnalyticsQuery Required: ITimeSeriesStore, IAuditLog	FR-002
Compliance & Audit Logger	Cross-Cutting Security	Captures all telemetry ingestion batches, breach events, caregiver acknowledgments, and administrative threshold edits into an immutable WORM ledger.	Provided: IAuditLog, IAuditQuery Required: IImmutableAuditStore, ICryptoHashChain	NFR-001, NFR-002

4. Architectural Pattern Selection & Justification

Architecture Selection Statement:

"We chose an Event-Driven Layered Microservices Architecture for the Remote Patient Vitals Alert & Monitoring System."

4.1 Candidate Architectural Patterns Evaluated

- Pattern 1 - Monolithic Architecture: Traditional single-deployable application. Rejected due to severe bottlenecks during peak telemetry ingestion (500+ streams) and single-point-of-failure risks in life-critical alerting.
- Pattern 2 - Traditional 3-Tier Client-Server Architecture: Synchronous request-response model. Rejected because polling for critical vital breaches introduces unacceptable alerting latency (>5s) and excessive server overhead.
- Pattern 3 - Pure Fine-Grained Microservices: High operational overhead and multi-hop network serialization latency that degrades deterministic sub-2s alert response times.
- Pattern 4 - Event-Driven Layered Microservices (CHOSEN): Combines asynchronous reactive telemetry stream processing (MQTT/WebSockets over TLS 1.3) with in-memory Redis state timers for fail-safe caregiver escalation, dedicated time-series persistence, and immutable cryptographic audit logging.

4.2 Quantitative Trade-Off Analysis Matrix

Architectural Quality Driver	Weight	Monolith	Client-Server	Pure Microservices	Event-Driven Layered (Chosen)
Anomaly Alert Latency (<2s, FR-001)	25%	3 (0.75)	2 (0.50)	3 (0.75)	5 (1.25)
High-Throughput Stream Ingestion (500+ Streams, NFR-001)	25%	2 (0.50)	2 (0.50)	4 (1.00)	5 (1.25)
Fail-Safe SLA Escalation (60s SLA, FR-005)	20%	3 (0.60)	2 (0.40)	4 (0.80)	5 (1.00)
HIPAA Security & Audit Immutability (NFR-002)	15%	2 (0.30)	3 (0.45)	4 (0.60)	5 (0.75)
Fault Isolation & System Uptime (99.99%)	15%	2 (0.30)	2 (0.30)	4 (0.60)	4 (0.60)
WEIGHTED OVERALL SCORE	100%	2.45 / 5.0	2.15 / 5.0	3.75 / 5.0	4.85 / 5.0

4.3 Specific Architectural Justifications

1. Scenario-Related Reason 1 (Deterministic Sub-2s Anomaly Detection): Event-driven reactive pipeline processes continuous telemetry in-memory without waiting for disk I/O, triggering emergency alerts within ~435ms of biometric breach.
2. Scenario-Related Reason 2 (Fail-Safe 60s Caregiver Escalation): The Alert Dispatcher leverages Redis key-expiration notifications to track the 60s acknowledgment SLA; if the primary caregiver fails to acknowledge, the system autonomously triggers the Backup Caregiver matrix.
3. Security Advantage (HIPAA & PHI Protection): End-to-end TLS 1.3 transport encryption, AES-256 field-level telemetry encryption, RBAC perimeter gating, and SHA-256 cryptographically chained audit logging prevent tampering and unauthorized PHI disclosure.
4. Performance Benefit (High-Throughput Concurrency): Asynchronous non-blocking I/O at the Ingestion Gateway effortlessly sustains 500+ simultaneous biometric streams while maintaining p99 processing latency under 50ms.

5. Requirements Traceability Matrix (Architecture to Lab 1)

Req ID	Requirement Summary	Realizing Component	Interface Contract	Acceptance Verification Mechanism
FR-001	Continuous vital evaluation & breach alert < 2s	Clinical Evaluation Engine	IVitalEvaluation	Automated stress pipeline triggers simulated critical breach (SpO2=86%); verifies alert dispatch in < 450ms.
FR-002	Secure live vitals dashboard	Patient App & Caregiver Portal	ITelemetryIngressAPI, ICaregiverAPI	Authenticated sessions render real-time telemetry streams updated at 1Hz with zero packet loss.
FR-003	Critical alert dispatch & caregiver acknowledgment	Alert Dispatcher & Escalation Matrix	IAlertNotification, IAlertAckService	Alert notification reaches caregiver within 3s; caregiver click sets state to Acknowledged and silences sirens.
FR-004	Dynamic clinical threshold configuration	Baseline Config Service	IConfigService, IThresholdRuleLookup	Caregiver updates patient limits via portal; evaluation engine applies new baseline within 10s without restart.
FR-005	Automated 60s timeout backup escalation	Alert Dispatcher & Escalation Matrix	IEscalationSLA, INotificationService	Simulated unacknowledged alert triggers Redis 60s timeout; verifies automatic dispatch to Backup Caregiver.
NFR-001	500 concurrent streams with 99.99% uptime	Telemetry Ingestion Engine	ITelemetryStream, ITimeSeriesStore	JMeter load test simulates 500 simultaneous streams over 60 min; confirms p99 latency < 500ms with 0 dropped frames.
NFR-002	HIPAA compliance, TLS 1.3, AES-256 & Audit Trail	Compliance & Audit Logger	IAuditLog, IImmutableAuditStore	Wireshark packet capture confirms TLS 1.3 encryption; audit ledger confirms immutable SHA-256 hash chains.

6. Security Architecture: STRIDE Threat Analysis

Threat Category	Healthcare Specific Attack Scenario	Architectural Countermeasure & Defense	Enforcing Component
Spoofing (S)	Rogue device impersonates patient wearable or unauthorized user impersonates caregiver.	Mutual TLS 1.3 device certificates, OAuth2 / OIDC authentication with RSA-256 signed JWT tokens.	API Gateway & Ingress Guard
Tampering (T)	Attacker alters biometric telemetry payloads or modifies historical vitals records.	HMAC-SHA256 payload signatures, database write-protection, and cryptographic WORM audit chaining.	Telemetry Ingestion & Audit Logger
Repudiation (R)	Caregiver denies receiving/ignoring critical vital breach notification.	Non-repudiable push delivery receipts and immutable timestamped acknowledgment audit logs.	Alert Dispatcher & Audit Logger
Information Disclosure (I)	Eavesdropping or leakage of confidential patient biometric telemetry.	Zero-Trust RBAC gating, TLS 1.3 in transit, and AES-256 envelope encryption at rest.	API Gateway & Persistence Tier
Denial of Service (D)	High-frequency telemetry floods overwhelm evaluation engine and delay alerts.	Token-bucket rate limiting at Ingress Gateway, non-blocking asynchronous event streaming.	API Gateway & Ingestion Engine
Elevation of Privilege (E)	Patient user attempts to access clinical threshold configuration endpoints.	Strict Role-Based Access Control (RBAC) enforced at Edge Gateway and service boundaries.	API Gateway & Baseline Config Service

7. Failure Modes & Latency Budget Analysis

7.1 Sub-2s Emergency Alert Latency Budget Breakdown (FR-001)

1. Edge Network Ingress & TLS Termination: 15 ms
2. Asynchronous Ingestion Gateway & Serialization: 25 ms
3. In-Memory Threshold Rule Evaluation: 10 ms
4. Alert Dispatcher Payload Construction & Redis State Lock: 35 ms
5. APNs / FCM Push & High-Priority SMS Gateway Delivery: 350 ms

TOTAL ESTIMATED END-TO-END ALERT LATENCY: 435 ms (Strictly adheres to the ≤ 2000 ms SLA constraint by a 4.5x margin)

7.2 Failure Mode Mitigations

- **Primary Caregiver Timeout:** Automated Redis countdown timer triggers Backup Caregiver escalation at 60s.
- **Sensor Stream Disconnection:** Ingestion engine flags connection timeout (>5s) and displays an active offline warning on the dashboard.
- **Ingestion Broker Failover:** Multi-node Kafka/RabbitMQ cluster ensures zero packet drop during single-node failure.